

Dr. Akshay Patil, PhD
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WORK EXPERIENCE

- **Senior SAR Specialist | Solafune Inc.** (Nov'2024- present)
- **Remote Sensing Scientist | AiDASH inc** (March'2021- Oct'2024)
 - Patented a technology and workflow for identification of hazard trees (submitted).
 - Developed a solution to access health of tree and detect wide scale tree mortality using multi-source remote sensing and machine learning.
 - Project management: Design and manage remote sensing projects by identifying requirements, developing project plans, overseeing data acquisition and processing, and ensuring project objectives were met.
 - Labelling and QC Workflow: Collaborate with the GIS team, ensuring adherence to best practices in data handling for machine learning/AI models, including labeling, multi-source image and shapefile processing, and mitigating potential errors.
 - Efficient Workflow Creation: Created and implemented an efficient delivery HILO workflow for the tree health product, optimizing the delivery process.
 - Data Sanity Checks: Evaluate data quality from third-party vendors and engage with potential collaborators and data providers for long-term partnerships.
 - Technical Due Diligence: Represented the R&D department in handling tech due diligence for Series B funding at Aidash.
 - Strategic Contribution to SaaS Vertical: Played a pivotal role in the development of the Intelligent Emission Management System (IEMS) SaaS vertical, leading to a strategic investment from Shell Ventures.
 - Wildfire Mapping and Risk Assessment: Created the solution for wildfire mapping and risk assessment, now an integral offering within Aidash's Climate Risk Intelligence System (CRIS) vertical.
- **Junior Research Fellow | Indian Institute of Technology, Bombay** (Aug'2015-Jul'2016)
 - Development of fully polarimetric SAR snow density retrieval algorithm for operational use in Indian Himalaya.
 - Development of bi-static radar remote sensing techniques for SWE retrieval.
- **Teaching Assistant | CSRE, IITB** (Aug'2016-Jul'2019)
 - Duties- Preparation of lecture notes, quiz material, conduct lab sessions and provide a guidance to undergraduates
- **Teaching Assistant | Monash University, Melbourne, Australia** (Aug'2019-Aug'2020)
 - Support professors on their daily teaching and laboratory activities (Evaluation of quiz papers, laboratory experiments).

• **Assistant Professor, E&TC | Sandip University**

(Aug'2014-Jul'2015)

- Teaching courses- Antenna and Wave Propagation, Electrostatic and Electromagnetism, Digital Signal Processing.

SCIENTIFIC CONTRIBUTIONS

- Developed MST Radar data processing chain in Matlab (data calibration, analytics, visualisation, simulations to identify meteor mass disintegration/fragmentation mechanism).
- Developed an algorithm that retrieves the snow parameters using single, dual, and full parametric SAR data at X- and C-band.
- Developed flood detection and oil spill detection algorithms using Sentinel-1 time series data.
- Evaluated capabilities of multi-source data (optical images spatial resolution of 50cm to 7cm) and time-series Sentinel-2/ PlanetScope for tree health assessment.
- Forest biomass retrieval using SAR and optical (stereo-optical and Lidar) data sources.
- Condition assessments of habitats - Wetlands, peatland and soil

FIELDS OF INTEREST AND EXPERTISE

- Snow parameter retrieval using SAR, forest/tree health assessment using optical remote sensing, forest disturbance mapping, precision agriculture, change detection using the EO satellites data (oil spill detection, flood detection, mapping wildfire burn scars).
- SAR and multispectral remote sensing
- Radar data processing (SAR and MST radars), ionospheric studies (centered around meteor phenomenon), image and signal processing, and electromagnetics.
- Machine learning, deep learning, and geospatial data science.
- InSAR, PolSAR data processing and applications.

TECHNICAL SKILLS

- **Script** (Python, Matlab, R, Javascript)
- **Tools** (ArcGIS Desktop, Quantum GIS, ERDAS Imagine, ClimateEngine, ENVI, PolSARpro, SARscape, SNAP ESA Toolbox, ISCE for InSAR processing, Google Earth Engine)
- **Drafting and Design** (Latex, Microsoft Word, Powerpoint and Excel, Lucid chart)
- **Source Control Tool** (GitHub, BitBucket)
- **SWork/ Project Management Tools** (Atlassian Jira and Confluence, Asana)

RESEARCH PUBLICATIONS

- **Akshay Patil**, G. Singh, C. Rüdiger, S. Mohanty, S. Kumar and Snehmmani, "A Novel Approach for the Snow Water Equivalent Retrieval Using X-Band Polarimetric Synthetic Aperture Radar Data," in IEEE Transactions on Geoscience and Remote Sensing, <https://ieeexplore.ieee.org/document/9177310>.
- **Akshay Patil**, G. Singh, and C. Rüdiger. "Retrieval of Snow Depth and Snow Water Equivalent using Dual Polarization SAR Data." in *MDPI Remote Sens*, 2020, <https://www.mdpi.com/2072-4292/12/7/1183>.
- **Akshay Patil**, S. Mohanty and G. Singh. "Snow depth and snow water equivalent retrieval using X-band PolInSAR data" in *Remote Sensing Letters* 2020, <https://www.tandfonline.com/doi/full/10.1080/2150704X.2020.1779373>.
- **Akshay Patil**, A. Malhotra, A.K. Patra, T.R. Prasad and J.D. Mathews (2014), "Evidence of meteoroid fragmentation in specular trail echoes observed using Gadanki MST radar", *Earth, Moon, and Planets*, Volume 114, issue 3-4, pp 89-90, <https://link.springer.com/article/10.1007/s11038-014-9456-4>.
- Closson, Damien, **Akshay Patil**, Mohamed Musthafa, Megan Gallagher, and Nitin Das. 2023. "Impact of Dead Sea Halo-Karst Development on an Earthen Dike Rehabilitation Project" *Geosciences* 13, no. 2: 42. <https://doi.org/10.3390/geosciences13020042>.
- Singh, G.; Lavrentiev, I.I.; Glazovsky, A.F.; **Akshay Patil**; Mohanty, S.; Khromova, T.E.; Nosenko, G.; Sosnovskiy, A.; Arigony-Neto, J. Retrieval of Spatial and Temporal Variability in Snowpack Depth over Glaciers in Svalbard Using GPR and Spaceborne POLSAR Measurements. *Water* 2020, 12, 21, <https://www.mdpi.com/2073-4441/12/1/21>.

CONFERENCE PROCEEDINGS

- **Akshay Patil**, G. Singh, S. Kumar, Sneh mani, D. Bandyopadhyay, B. R. Nela, M. Musthafa, and S. Mohanty, "Snow Characterization and Avalanche Detection in the Indian Himalaya", IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium, Hawaii, USA, 2020, <https://ieeexplore.ieee.org/document/9324621>.
- **Akshay Patil**, G. Singh, and C. Rüdiger, "Snow water equivalent and snow depth estimation in the Himalayas," Water future conference, Bengaluru, India, 2019.
- **Akshay Patil**, G. Singh and C. Rüdiger, "A Novel Approach for The Retrieval Of Snow Water Equivalent Using SAR Data," IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium, Yokohama, Japan, 2019, pp. 3233-3236. <https://ieeexplore.ieee.org/document/8900565>.
- **Akshay Patil**, G. Singh and C. Rüdiger, "Estimation of Snow Water Equivalent Using Sentinel SAR Data in the Indian Himalaya," IGARSS 2018 – 2018 IEEE International Geoscience and Remote Sensing Symposium, Valencia, 2018, pp. 2789-2792, <https://ieeexplore.ieee.org/abstract/document/8518432>.
- **Akshay Patil**, A. Malhotra, A.K. Patra, T.R. Prasad, "Specular meteor trail observations using MST Gadanki radar," RCRS-2014, Pune, 2014.

- **Akshay Patil**, G. Singh, S. Kumar, Snehmani, and C. Rüdiger, "Snow surface characterization using Polarimetric SAR (PolSAR) data," 36th International Geological Congress 2020, Delhi, India.
- D. Bandyopadhyay, G. Singh, G. Dasaundhi, B. R. Nela, **Akshay Patil**, and S. Mohanty, "Surging Glacier Dynamics in Tarim Basin Using SAR Data", IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium, Hawaii, USA, 2020, <https://ieeexplore.ieee.org/document/9323095>.
- B. R. Nela, G. Singh, D. Bandyopadhyay, **Akshay Patil**, S. Mohanty, M. Musthafa, and G. Dasundhi, "Estimating Dynamic Parameters of Bara Shigri Glacier and Derivation of Mass Balance from Velocity", IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium, Hawaii, USA, 2020, <https://ieeexplore.ieee.org/abstract/document/9323152>.
- M. Musthafa, G. Singh, **Akshay Patil**, and B. N. Raju, "Forest Above Ground Biomass Estimation Using Multi-Sensor Geostatistical Approach", IGARSS 2020 - 2020 IEEE International Geoscience and Remote Sensing Symposium, Hawaii, USA, 2020, <https://ieeexplore.ieee.org/abstract/document/9323642>.
- R. Gautam and **Akshay Patil**, "Modeling and control of joint angles of a biped robot leg using PID controllers," 2015 IEEE International Conference on Engineering and Technology (ICETECH), Coimbatore, 2015, pp. 1-5, <https://ieeexplore.ieee.org/document/7275042>.
- A. Malhotra, **Akshay Patil**, B. Gao, A. K. Patra, T. R. Prasad, and J. D. Mathews, "High altitude meteor observations using Gadanki and Jicamarca VHF radars", RCRS-2014, Pune, 2014.
- D. Bandyopadhyay, B. R. Nela, G. Singh, A. V. Kulkarni, **Akshay Patil**, "Contrasting Mass Balance changes in the Himalaya estimated using SAR remote sensing," 36th International Geological Congress 2020, Delhi, India
- B. R. Nela, G. Singh, D. Bandyopadhyay, and **Akshay Patil**, "Ice thickness distribution of Chandra and Bhaga basin glaciers derived from DInSAR based velocity map using Laminar flow law", 36th International Geological Congress 2020, Delhi, India

Grants

- IEEE Geoscience and Remote Sensing Society Travel Grant, IEEE IGARSS-2018 (**Valencia, Spain**) July 2018/
- IEEE Geoscience and Remote Sensing Society Travel Grant, IEEE IGARSS-2019 (**Yokohama, Tokyo, Japan**) July 2019.
- Monash University Travel Grant for Water Future Conference-2019 (Bangaluru, India)

GUEST LECTURES AND WORKSHOPS

- **IIT Bombay, IEEE GRSS Chapter, Mumbai**, "Solving Industry Use Cases with AI and Remote Sensing: Applications that Matter in the Real World Scenario"
- **IIT Indore, IEEE GRSS Chapter**, "Revolutionizing Real-time Industry Applications through Remote Sensing"
- **IIT Indore**, "Drone Application in Remote Sensing"
- **VIT, Vellore**, "Application of Remote Sensing, GIS and AI in Industries: The Real-time Scenarios"

- **Let's Talk Spatial, GalaxEye, Bangalore,** *"Real-Time Insights from the Sky: Unleashing the Potential of the SAR"*
- **Workshop WRSC-2023, Mumbai,** *"Remote Sensing of Cryosphere"*

RESEARCH PROJECTS

- **NASA international space apps challenge: COVID19-COARSE Analysis,** *(Region-South and Central Asia)* (May'2020)
 - <https://covid19.spaceappschallenge.org/challenges/covid-challenges/quiet-planet/teams/coarse-analysis/project>
- **PAZ mission: Snow water equivalent retrieval using dual polarization SAR data.,** *(PhD Project, Co-investigator)* (2019-2020)
 - Cascade (merge or harmonise) the TerraSAR-X time series data with PAZ time-series for snow studies.
- **BRICS Project: Collaboration on Glaciers and Snow/Firn Characteristics Variability Studies in Polar and Subpolar Regions)** *(PhD Project, Co-investigator)* (2018-2019)
 - Retrieval of Spatial and Temporal Variability in Snowpack Depth over Glaciers in Svalbard Using GPR and Spaceborne POLSAR Measurements biomass estimation.
- **Analysis of specular meteor trails and high altitude radar meteor data using MST radar, Gadanki)** *(MTech Project, Co-investigator, NARL(ISRO)/Penn State University)* (2012-2014)
 - MST radar phase calibration using in-phase and quadrature components.
 - Simulation of meteor mass disintegration mechanism
 - Finding the speed and mass loss mechanism of meteor.

Field Campaigns

- Snow field (2020)- The field campaign with joint collaboration by SASE – IIT Bombay in Himachal Pradesh, India. Aimed for Snow data collection in synchronous with Satellite (PAZ, Sentinel-1A, Sentinel-2) overpasses.
- Snow field (2018) - The field campaign with joint collaboration by SASE – IIT Bombay in Himachal Pradesh, India. Aimed for Snow collection in synchronous with Satellite (Radarsat-2, Sentinel-1A, Sentinel-2) overpasses
- Snow field (2016)- The field campaign with joint collaboration by SASE – IIT Bombay in Himachal Pradesh, India. Aimed for Snow collection in synchronous with Satellite (Radarsat-2, Sentinel-1A, Sentinel-2) overpasses.
- Forestry field (2016) - Field collection in Chandrapur, Maharashtra, India to develop and validate forest parameter and biomass applications.
- Forestry field (2015)- Field and Satellite data collection in Thanu and Barkot, Uttarakhand, India to develop and validate forest parameter and biomass applications.

Synergistic Activity, Professional Membership, and Certifications

- Journal Reviewer:
 - American Journal of Electromagnetic and Applications
 - International Journal of Remote Sensing (Taylor & Francis)
 - IEEE transactions on geoscience and remote sensing
 - Journal of Indian Society of Remote Sensing
 - Remote Sensing Letters
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing
 - Photonirvachak Journal of Indian Society of Photo-interpretation and Remote Sensing
- IEEE Geoscience and Remote Sensing Society (S'17) | Student Member
- Disaster Management | Certification Course
- Basics of JavaScript Web Apps | ESRI
- Getting Started with GIS | ESRI
- Python for Everyone | ESRI
- Hyper EDU certificate | EO College
- Python OOP: Four Pillars of OOP in Python 3 | Udemy
- GEOINT Professional- GIS & Analysis Tools (CGP-G)
- Microwave Remote Sensing Workshop-2015 | CSRE-IIT Bombay
- Organizer of 'Circuit Debugging', a state level competition in 'MET-EORITE 2010', Nashik, India.
- Second Prize in "Circuit Junkies", a state level competition, Pune, India.
- Techfest-Techconnect 2020 (Exhibitor), IIT Bombay

DECLARATION AND SIGN OFF

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.

Dr. Akshay Patil
DOB: 5 Oct' 1989